

FrontierMem: Qwen2.5-1.5B Ablation Study

Learning When a User Preference Should Apply, Be Ignored, or Trigger Clarification

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Executive result. On 72 balanced FrontierMem cases, the best configuration was **rule-extracted memory + LLM decision** with **70.83% accuracy** and **68.45% Macro-F1**. The modular **LLM memory + rule policy** reached 66.67%, while the full two-stage LLM pipeline collapsed to **IGNORE on all 72 examples**, giving 33.33% accuracy. The experiment therefore supports a modular design and identifies the final LLM decision/response stage as the immediate integration bottleneck.

1 Goal and Experimental Setup

FrontierMem treats personalization as an *applicability-boundary* problem rather than a flat fact-retrieval problem. For each query, the system must choose one of three actions: **APPLY** a relevant preference, **IGNORE** an out-of-scope preference, or **CLARIFY** when a material boundary condition is missing.

The attached Kaggle notebook evaluated **Qwen2.5-1.5B-Instruct** on a controlled 72-example subset: 24 examples per action, 12 scenario families, and four equally represented domains (education, food, travel, and writing). The environment used Transformers 4.57.6, Accelerate 1.14.0, PyTorch 2.10.0+cu128, and an active CUDA runtime.

Table 1: Four modular evaluation settings.

Mode	Memory source	Final action source
Gold memory + LLM	Ground-truth structured memory	Label-only LLM classifier
Rule memory + LLM	Deterministic rule extractor	Label-only LLM classifier
LLM memory + rule	LLM-generated structured memory	Deterministic applicability policy
Full LLM pipeline	LLM-generated structured memory	LLM JSON decision and response

2 Main Results

Table 2: Complete aggregate results. All values are percentages. Twin metrics are based on only three different-gold twin pairs and should be treated as diagnostic rather than conclusive.

Mode	Acc.	Bal. Acc.	Macro-F1	Inside	Outside	Near	Overgen.	Unnec. Clar.	Twin exact
Gold memory + LLM	62.50	62.50	60.04	100.00	29.17	58.33	58.33	6.25	66.67
Rule memory + LLM	70.83	70.83	68.45	100.00	33.33	79.17	54.17	6.25	100.00
LLM memory + rule	66.67	66.67	65.35	83.33	37.50	79.17	12.50	33.33	33.33
Full LLM pipeline	33.33	33.33	16.67	0.00	100.00	0.00	0.00*	0.00*	0.00

*The zero error rates of the full pipeline are not positive evidence: the model never applied a preference or asked for clarification because it predicted IGNORE for every example.

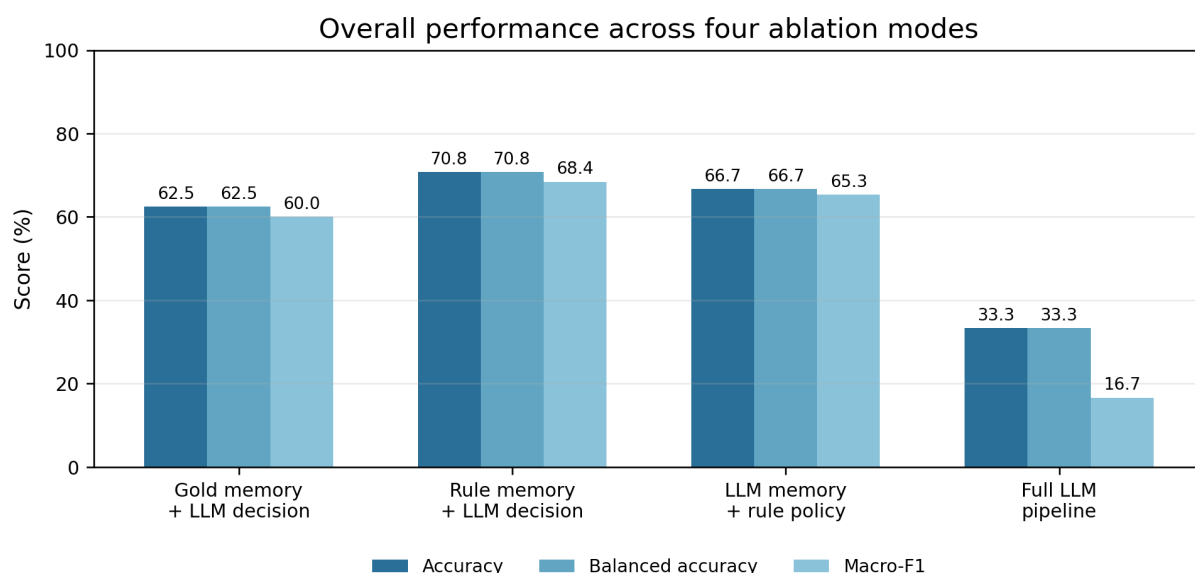


Figure 1: Overall accuracy, balanced accuracy, and Macro-F1.

The best overall mode used **rule-extracted memory with an LLM decision**. However, its 42 APPLY predictions versus only 8 IGNORE predictions reveal a strong application bias. The **LLM-memory + rule-policy** mode was slightly lower overall but produced the most balanced operational trade-off: it reduced over-generalization to 12.5%, at the cost of more unnecessary clarification (33.3%).

3 Boundary Behaviour and Class Bias

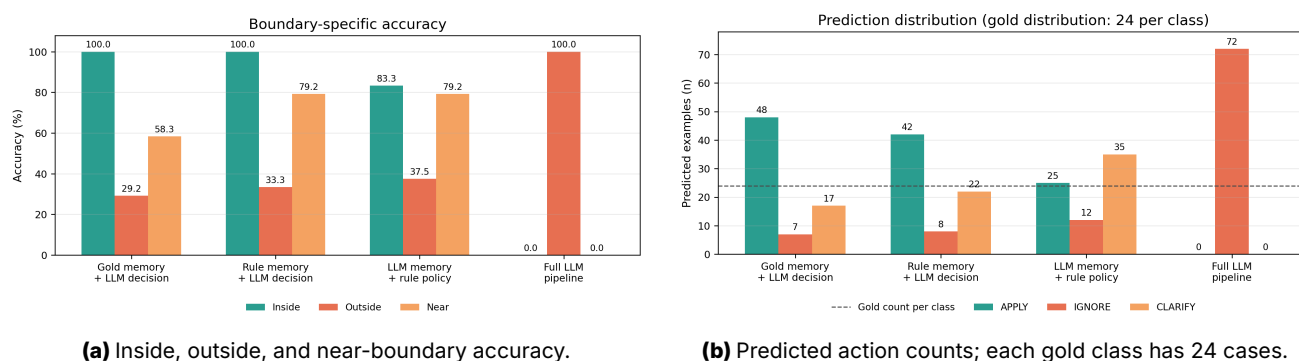
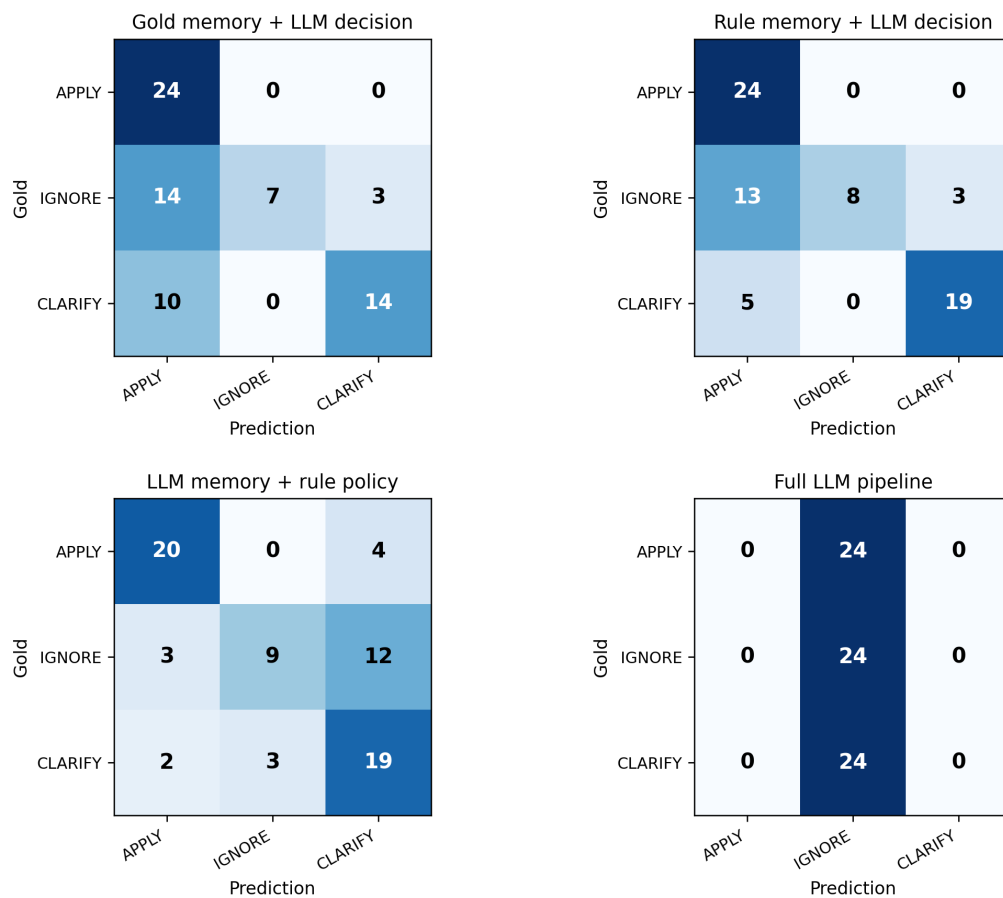


Figure 2: Boundary-level performance and prediction bias.

Across the three non-collapsed modes, **outside-boundary decisions were consistently the hardest** (29.17–37.50% accuracy). Gold-memory and rule-memory LLM decisions over-applied preferences: APPLY recall was 100%, but IGNORE recall was only 29.17% and 33.33%, respectively. In contrast, the rule policy on LLM memories shifted errors toward CLARIFY, producing 35 clarification decisions and reducing direct overgeneralization.

Confusion matrices (counts; 24 examples per gold class)

**Figure 3:** Confusion matrices for all four modes. The full pipeline shows complete class collapse to IGNORE.

4 Memory Extraction Quality

The same deterministic LLM memory extraction was used in both `llm_memory_rule` and `full_llm`; their structured memories matched row-for-row. Extraction was strong for the broad applicability attributes but weak for ontology-specific fields.

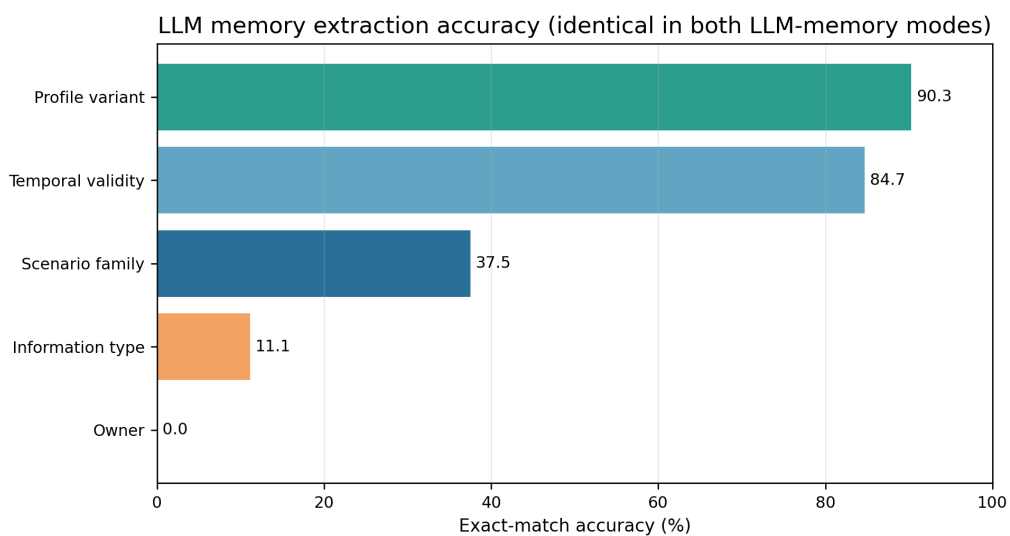
**Figure 4:** Exact-match accuracy of extracted memory fields.

Table 3: Memory extraction results on 72 examples.

Field	Correct	Accuracy	Interpretation
Profile variant (stable/scoped)	65/72	90.28%	Strong high-level boundary detection
Temporal validity	61/72	84.72%	Strong persistent/context-dependent distinction
Scenario family	27/72	37.50%	Frequent family collapse and cross-domain confusion
Information type	8/72	11.11%	Mostly collapsed to <code>temporary_constraint</code>
Owner	0/72	0.00%	Produced free-form values such as session IDs or <code>none</code>

This pattern is important: the model can often infer *whether* a preference is stable or contextual, but it does not reliably follow the closed schema for *who owns the information, what type it is, or which family it belongs to*. Supervised schema learning and constrained field vocabularies are therefore necessary.

5 Why the Full Pipeline Failed

Full LLM pipeline breakdown

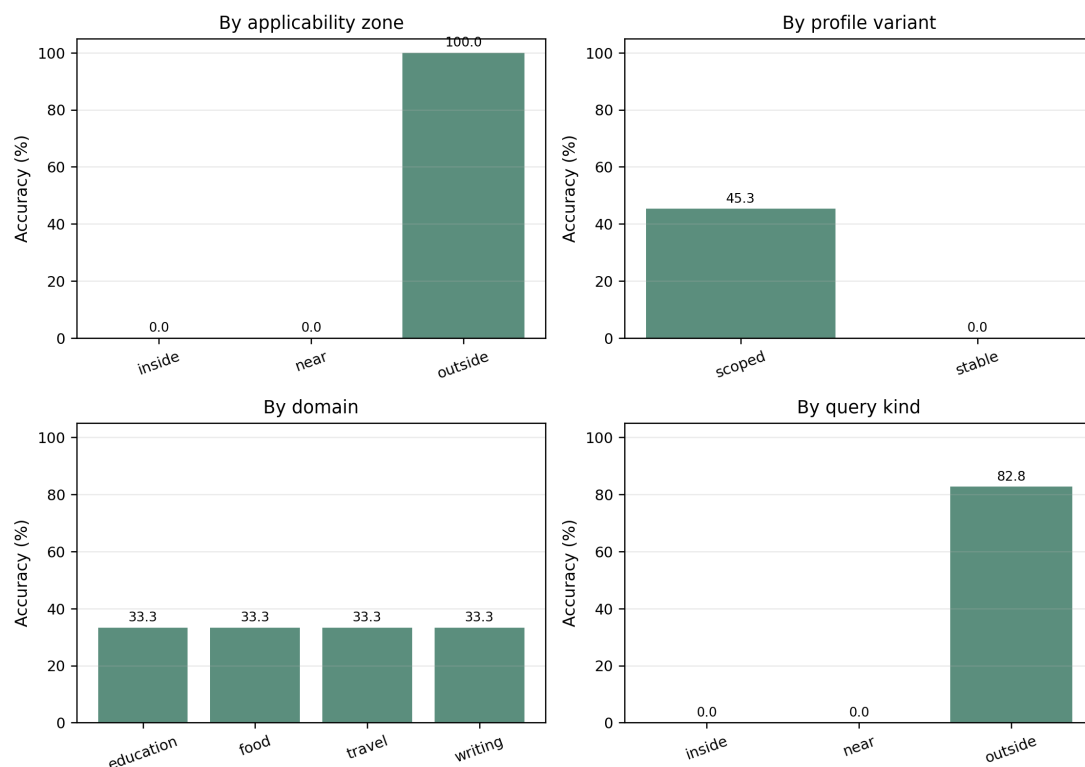


Figure 5: Breakdown of the full LLM pipeline. Uniform 33.3% domain accuracy is a consequence of predicting the same label in every domain.

The full pipeline generated valid JSON for all 72 memory outputs and all 72 decision outputs, with **zero parser fallbacks**. Therefore, this was not a loading or parsing failure. It was a semantic decision-stage collapse:

Table 4: Full-pipeline diagnostic indicators.

Indicator	Result	Meaning
Predicted action	72/72 IGNORE	Complete action collapse
Predicted relation	42 mismatch, 30 unknown	No query was treated as a positive match
Decision confidence	0.0 for 72/72	The structured decision stage was not calibrated
Missing generated response	31/72	Response generation was incomplete despite valid JSON
Zone accuracy	Inside 0%, Outside 100%, Near 0%	Only the IGNORE gold class was recovered
Stable profiles	0/19 correct	Persistent preferences were always suppressed
Parser fallback	0/72	Errors are semantic, not syntactic

The strongest diagnostic comparison is between **LLM memory + rule policy** and the **full LLM pipeline**. They used exactly the same extracted memories, yet replacing the LLM decision stage with a deterministic policy increased accuracy from 33.33% to 66.67%. This isolates the immediate catastrophic failure to the integrated LLM decision/response prompt rather than to memory extraction alone. At the same time, the 62.50% result with gold memory shows that the underlying decision capability is still imperfect even when the memory is correct.

6 Representative Outputs

Table 5: Representative full-pipeline cases.

Case	Query (abridged)	Gold	Pred.	Diagnosis
Scoped education detail, outside	"I only need a quick correctness check on a familiar exercise."	IGNORE	IGNORE	Correct action, although the extracted family was wrong and the reason stated that a fact was missing.
Stable writing directness, inside	"Write a compassionate message to someone facing bad news."	APPLY	IGNORE	Compassion was incorrectly treated as a mismatch rather than a compatible context for a stable directness preference.
Scoped quick meals, near	"I have not said whether I am still busy or finished with work."	CLARIFY	IGNORE	A missing scope condition should have triggered clarification, but the model suppressed the preference instead.
Stable vegetarian preference, inside	"Plan dinner for my sister, who still follows a vegetarian diet."	APPLY	IGNORE	The action stage ignored a clearly applicable dietary preference despite a largely correct memory.

7 Decisions, Limitations, and Next Steps

Decisions supported by the experiment

- Keep **memory extraction and action selection modular**. The hybrid modes are currently much more reliable than the end-to-end two-call pipeline.
- Replace the full JSON decision generator with a **label-only classifier** (or a deterministic relation-to-action policy), then generate the natural-language response in a separate third step.
- Enforce **closed vocabularies** for owner, information type, family, relation, and action; reject or normalize out-of-schema values before evaluation.
- Fine-tune on counterfactual twins with explicit stable/scoped, inside/outside/near, and owner/type supervision. Prioritize outside-boundary examples because IGNORE recall remains below 38% in all non-collapsed modes.
- Expand twin evaluation beyond the current three different-gold pairs and run multiple seeds before making comparative claims.

The dataset is controlled and synthetic, and the rule extractor is aligned with its templates; consequently, the 70.83% best score should not be interpreted as real-world generalization. The surprising advantage of rule memory over gold memory also indicates representation sensitivity: a compact, cue-aligned memory can be easier for a small LLM to classify than a richer ground-truth record. This requires a dedicated memory-format ablation.

A Qwen2.5-3B gold-memory run was initiated in the notebook, but only a partial model download was recorded and no evaluable metrics were produced; it is therefore excluded from the quantitative comparison. The next 3B run should be performed only after fixing the decision interface, otherwise model scaling may reproduce the same integration collapse.

Conclusion

Qwen2.5-1.5B shows useful but incomplete FrontierMem capability. It recognizes stable versus scoped memories and temporal validity reasonably well, and it can reach 62–71% accuracy in modular settings. However, it struggles to ignore out-of-scope preferences, does not reliably obey the memory ontology, and completely collapses when memory extraction, JSON action selection, explanation, and response generation are combined. The current evidence favors a **structured hybrid pipeline**: constrained memory extraction, explicit applicability policy, and separate response realization, followed by supervised fine-tuning and broader counterfactual evaluation.

Artifacts analyzed: the submitted Kaggle notebook and the complete Qwen2.5-1.5B result package, including four prediction files, four aggregate metric files, confusion matrices, extraction metrics, per-group breakdowns, parser statistics, and representative examples. Full per-example outputs remain available in the accompanying CSV files.